

ASSIGNMENT-4

1. Write a program in C to find the sum of the series $1!/1+2!/2+3!/3+4!/4+5!/5$ using the function

```
#include <stdio.h>

int sum(int num){
    if(num==0 || num==1){
        num=1;
    }
    else
        return num*sum(num-1);
}

int main()
{
    int num;
    int ans =0;
    printf("enter the number=")
    scanf("%d",&num);
    for(int i=1;i<=num;i++){
        ans=sum(i)/i;
        ans+=ans;
    }
    printf("ans=%d\n",ans);

    return 0;
}
```

2. Write a program in C to convert a decimal number to binary number using the function

```
#include <stdio.h>

void convert(int n){
    int binaryNum[32];
```

```

// counter for binary array
int i = 0;
while (n > 0) {
    // storing remainder in binary array
    binaryNum[i] = n % 2;
    n = n / 2;
    i++;
}

// printing binary array in reverse order
for (int j = i - 1; j >= 0; j--)
    printf("%d", binaryNum[j]);

}

int main()
{
    int num;
    scanf("%d",&num);
    convert(num);

    return 0;
}

```

OUTPUT:

67

1000011

3. Write a program in C to print the first 50 natural numbers using recursion.

```

#include <stdio.h>

int natural(int n){
    if(n==1){
        n=1;
        printf("%d",n);
    }
}

```

```

    }
    else{
        printf("%d\t",n);

        return natural(n-1)+1;
    }
}
int main()
{
    int num;
    scanf("%d",&num);
    natural(num);
    // printf("%d",natural(num));

    return 0;
}

```

OUTPUT:

5

5 4 3 2 1

4. Write a program in C to find GCD of two numbers using recursion.

```

#include <stdio.h>

int hcf(int n1, int n2);

int main() {
    int n1, n2;
    printf("Enter two positive integers: ");
    scanf("%d %d", &n1, &n2);
    printf("G.C.D of %d and %d is %d.", n1, n2, hcf(n1, n2));
    return 0;
}

```

```

int hcf(int n1, int n2) {
    if (n2 != 0)
        return hcf(n2, n1 % n2);
    else
        return n1;
}

```

OUTPUT:

Enter two positive integers: 6 7

G.C.D of 6 and 7 is 1.

5. Write a program in C to calculate the power of any number using recursion.

```

#include <stdio.h>

int power(int n1, int n2);

int main() {
    int base, a, result;
    printf("Enter base number: ");
    scanf("%d", &base);
    printf("Enter power number(positive integer): ");
    scanf("%d", &a);
    result = power(base, a);
    printf("%d^%d = %d", base, a, result);
    return 0;
}

```

```

int power(int base, int a) {
    if (a != 0)
        return (base * power(base, a - 1));
    else
        return 1;
}

```

6. Write a program in C to find the sum of digits of a number using recursion

```
#include <stdio.h>
```

```
int sum (int a);
```

```
int main()
```

```
{
```

```
    int num, result;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &num);
```

```
    result = sum(num);
```

```
    printf("Sum of digits in %d is %d\n", num, result);
```

```
    return 0;
```

```
}
```

```
int sum (int num)
```

```
{
```

```
    if (num != 0)
```

```
    {
```

```
        return (num % 10 + sum (num / 10));
```

```
    }
```

```
    else
```

```
    {
```

```
        return 0;
```

```
    }
```

```
}
```

OUTPUT:

2022

6

7. Write a program in C to find the Hailstone Sequence of a given number upto 1. Hailstone Sequence:

```
#include <stdio.h>
```

```
int HailstoneNumbers (int N)
```

```
{
```

```
    int c;
```

```
    printf ("%d\t", N);
```

```
    if (N == 1 && c == 0)
```

```
    {
```

```
        return c;
```

```
    }
```

```
    else if (N == 1 && c != 0)
```

```
    { c++;
```

```
        return c;
```

```
    }
```

```
    else if (N % 2 == 0)
```

```
    {
```

```
        c++;
```

```
        HailstoneNumbers (N / 2);
```

```
    }
```

```
    else if (N % 2 != 0)
```

```
    { c++;
```

```
        HailstoneNumbers (3 * N + 1);
```

```
    }
```

```
}
```

```
int main ()
```

```
{
```

```
    int N;
```

```
scanf ("%d", &N);  
  
int x;  
  
HailstoneNumbers (N);  
  
return 0;  
  
}
```

OUTPUT:

5

5 16 8 4 2 1